

Chapter 2 Questions

1. Potential outcomes notation

- a. $Y_i(0)$ is the outcome person i would have without receiving the treatment.
 - b. D_i represents the random treatment assignment before the assignment has taken place. The d_i represents the assignment that occurred in the actual experiment.
- $Y_i(0)$ means the outcome under control for people assigned to treatment. This outcome is not observed because these people received treatment.
- c. $Y_i(0)$ is person i 's possible outcome without treatment, regardless of their actual group. $Y_i(0) / D_i = 0$ is the outcome without treatment among people assigned to the control group.
 - d. $Y_i(0) / D_i = 1$ is the outcome treated people would have had without treatment. $Y_i(0) / D_i = 0$ is the outcome of people assigned to control, so it is observed.
 - e.
 - f. With random assignment, people are placed into treatment and control by chance. Therefore, the two groups should have the same average control outcome in expectation, making the selection-bias term zero.

2. Student newspaper

- a. The letter may violate excludability because students could change their political interest after learning the purpose of the study. Their behaviour might therefore be affected by both the newspaper and the information in the letter.
- b. Sharing newspapers may violate non-interference because control students could read newspapers belonging to treated students. One student's treatment would then affect another student's outcome.

3. Prison inmates

- a. Reading was not randomly assigned. Prisoners chose how much to read, and prisoners who read more may already be calmer or less violent. Therefore, the study shows an association but does not prove that reading causes less violence.
- b. The experiment changes more than reading. Treated prisoners also spend time in the library, sit in special carrels, and are separated from their usual environment. These changes could reduce violence even if reading itself has no effect, threatening excludability.
- c. Non-interference means that assigning one prisoner to the reading treatment should not change another prisoner's violent behaviour.
- d.